CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 2: Algorithms, Programming and Logic | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 09: DATABASES

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/22/M/J/24 Q5 (2024 Series)] [Total: 7 marks]

(a) Define the following database terms:

(i) Field

(ii) Record

(iii) Primary Key **[3 marks]**

(b) A database table named STUDENTS contains fields: StudentID (Text), FirstName (Text), LastName (Text), Age (Integer), Grade (Text).

Write an SQL query to display the FirstName and LastName of all students who are older than 14, sorted alphabetically by LastName in ascending order. **[4 marks]**

✓ Marking Scheme / Model Answer:

(a) (i) Field: Single column/attribute [1 mark]. (ii) Record: Single complete row of entity data [1 mark]. (iii) Primary Key: Unique identifier for each record [1 mark].

(b) SQL Query:

```
SELECT FirstName, LastName [1 mark]
FROM STUDENTS [1 mark]
WHERE Age > 14 [1 mark]
ORDER BY LastName ASC; [1 mark]
```

Question 2 [2210/22/M/J/23 Q5 (2023 Series)] [Total: 6 marks]

(a) A database table CARS has fields: CarID, Manufacturer, Model, Year, Price, InStock (Boolean).

Write an SQL statement to display the Manufacturer, Model, and Price of all cars where Year is 2020 or newer and InStock is TRUE, sorted with the highest Price first. **[4 marks]**

(b) Write an SQL query to find the average price (AVG) of all cars made by "Toyota". **[2 marks]**

✓ Marking Scheme / Model Answer:

(a) SQL Statement [4 marks]:

```
SELECT Manufacturer, Model, Price [1 mark]
FROM CARS [1 mark]
```

WHERE Year >= 2020 AND InStock = TRUE [1 mark]
ORDER BY Price DESC; [1 mark]
(b) SQL Statement [2 marks]:
SELECT AVG(Price)
FROM CARS
WHERE Manufacturer = 'Toyota';

Question 3 [2210/21/M/J/24 Q4 (2024 Series)] [Total: 5 marks]

(a) A database table BOOKS has fields: BookID, Title, Author, Price, StockQuantity.

Write an SQL statement to calculate the total number of books in stock (sum of StockQuantity). **[2 marks]**

(b) Write an SQL statement to count how many books have a Price less than 15.00. **[3 marks]**

✓ Marking Scheme / Model Answer:

(a) SELECT SUM(StockQuantity) FROM BOOKS; [2 marks]
(b) SELECT COUNT(BookID) FROM BOOKS WHERE Price < 15.00; [3 marks]
